

Prescribing every successive L^1 -minimum of a coefficient lattice

Mixed-radix realization, rigid minimizing bases,
and exact iterated-sumset growth

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Abstract

Let $\mathcal{L}(A)$ be the integer vectors perpendicular to both the all-ones vector and a vector listing the distinct integers of a finite set A . We prove that every nondecreasing list of even integers at least four occurs as the complete list of successive L^1 -minima of such a lattice, with exactly two more elements in A than entries in the list. For prescribed half-minima $2 \leq h_1 \leq \dots \leq h_r$, an explicit witness is

$$A = \{0, 1, h_1, h_1 h_2, \dots, h_1 \cdots h_r\}.$$

A stronger theorem permits the multipliers in any order. Its proof gives an integral basis of carry relations and bounds each basis coordinate of every short vector. It determines the entire subgroup generated by short relations and all successive minimizing bases. This resolves the substantive realization conjecture following Lemma 1 of O'Bryant's *On Nathanson's Triangular Number Phenomenon*, while correcting its impossible endpoint $h_1 = 1$. We also obtain unique mixed-radix normal forms, a complete-intersection presentation, an exact generating function for every iterated sumset, and the sharp onset of linear growth. All arguments are self-contained; exact-arithmetic verification programs and their outputs accompany the article.

Status. This is an AI-assisted mathematical research draft, not an independently refereed or proof-assistant-verified paper. The all-parameter claims are supported by the proofs below, not by extrapolation from tests. The source still states the target as a conjecture in the version inspected; no later solution was located in the searches recorded with this package. That search does not establish priority.

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1 The problem and the precise claim

The inverse problem studied here asks how freely one can prescribe the sizes at which linearly independent additive relations first appear in a set of integers. A relation between two multisets of the same cardinality is encoded by subtracting their multiplicity vectors. Its L^1 -norm is twice the number of terms remaining on either side after cancellation. Prescribing successive minima therefore means prescribing successive independent collision thresholds, not merely finding one short relation.

The explicit target is the unnumbered conjecture immediately after Lemma 1 in Section 1.1, page 2, of O’Bryant’s paper [1]. That lemma handles the first two minima; the following conjecture asks for a full prescribed list. The displayed lower endpoint is $h_1 = 1$. Section 2 proves that this endpoint cannot occur for a set of distinct integers. The realization question on its admissible domain is:

Given $r \geq 1$ and integers $2 \leq h_1 \leq \dots \leq h_r$, is there an $(r + 2)$ -element set of integers whose coefficient lattice has successive L^1 -minima $2h_1, \dots, 2h_r$?

The answer is affirmative. We do not stop at the endpoint obstruction: we construct a witness for *every* admissible list.

1.1 What is and is not being resolved

The central result is a full realization theorem, not a statement about a fixed number of minima, a generic parameter range, or a bounded search. Repeated minima and arbitrarily large gaps between consecutive minima are allowed. The witness is normalized by $\min A = 0$ and $1 \in A$. Its lattice has a genuine $\mathbb{Z}$ -basis attaining all the minima.

The additional sumset formulas apply to the constructed multiplicative-chain sets. They do not classify all sets with the same minima, nor all attainable sumset cardinalities. The larger range-of-sizes problem is discussed, for example, in Nathanson’s work [2]. Section 8 gives two sets with the same complete minimum list but different later sumset sizes. Thus the special-family formulas must not be promoted to formulas for arbitrary coefficient lattices.

1.2 Selection and exclusion audit

Before selecting a problem, a single call to an operating-system-backed random number generator selected entry 1 from a fixed catalogue of 96 areas: *additive combinatorics*. The complete catalogue and the recorded result are included in `data/area_selection.json`; the catalogue is also printed in Appendix A. There was no redraw.

All 71 entries in the supplied `manifest(1).tex` were inspected as exclusions. None describes this coefficient-lattice realization problem. The comparison uses the manifest’s descriptions; it is not a re-verification of the proofs in those other packages. The target and the distinction from the excluded topics are recorded in `notes/exclusion_audit.md`.

2 Coefficient lattices and necessary conditions

Fix a finite integer set

$$A = \{a_0 < a_1 < \dots < a_{k-1}\}, \quad k \geq 2.$$

Coordinates are numbered from zero throughout this article.

Definition 2.1. The *coefficient lattice* of A is

$$\mathcal{L}(A) = \left\{ c \in \mathbb{Z}^k : \sum_{i=0}^{k-1} c_i = 0, \quad \sum_{i=0}^{k-1} a_i c_i = 0 \right\}. \quad (2.1)$$

For $c \in \mathcal{L}(A)$ put $c_i^+ = \max(c_i, 0)$, $c_i^- = \max(-c_i, 0)$, and

$$d(c) = \sum_i c_i^+ = \sum_i c_i^- = \frac{1}{2} \|c\|_1. \quad (2.2)$$

For $1 \leq j \leq k-2$, its j th successive minimum is

$$\lambda_j(A) = \min \left\{ T \geq 0 : \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \{ c \in \mathcal{L}(A) : \|c\|_1 \leq T \} \geq j \right\}. \quad (2.3)$$

The two rows defining (2.1) are independent because the a_i are not all equal. Their rational kernel has dimension $k-2$. A rational basis can be multiplied by denominators to give independent integer vectors, so $\mathcal{L}(A)$ has rank $k-2$. Hence all the minima in (2.3) exist. Using $\mathbb{R}$ -linear instead of $\mathbb{Q}$ -linear independence gives the same definition for these rational vectors.

Proposition 2.2 (The complete elementary obstruction). *Every successive minimum of $\mathcal{L}(A)$ is an even integer at least four.*

Proof. Equation (2.2) makes every lattice norm an even integer. If a nonzero vector had degree one, its positive and negative parts would each be a coordinate unit vector. Thus it would equal $e_i - e_j$ for $i \neq j$. The second relation in (2.1) would imply $a_i = a_j$, contrary to distinctness. A nonzero relation therefore has degree at least two. $\square$

Consequently, the conjecture with the literal endpoint $h_1 = 1$ is false. Replacing that endpoint by two is necessary, and the next section proves that it is sufficient. The adjacent two-minimum lemma in the source already uses the endpoint two [1]; the endpoint discrepancy is stated explicitly here rather than silently repaired.

Translation and nonzero scaling do not change the lattice: if $b_i = u + va_i$ with $v \neq 0$, then, on $\sum c_i = 0$, $\sum b_i c_i = v \sum a_i c_i$. A negative scaling merely reverses the increasing coordinate order. The construction below therefore also gives strictly positive witnesses after adding one to every element.

3 A full realization theorem

It is useful to prove more than the ordered-parameter statement. Let $q_1, \dots, q_r \geq 2$ be integers in *any* order. Define

$$a_0 = 0, \quad a_1 = 1, \quad a_{j+1} = q_j a_j \quad (1 \leq j \leq r), \quad A(q) = \{a_0, \dots, a_{r+1}\}. \quad (3.1)$$

The elements are strictly increasing. Write

$$v_j = (q_j - 1)e_0 - q_j e_j + e_{j+1} \quad (1 \leq j \leq r). \quad (3.2)$$

This is the carry relation

$$\underbrace{a_j + \dots + a_j}_{q_j \text{ terms}} = a_{j+1} + \underbrace{0 + \dots + 0}_{q_j - 1 \text{ terms}}.$$

Theorem 3.1 (Mixed-radix spectrum theorem). *For $A(q)$ in (3.1), the following statements hold.*

(i) *The vectors $v_1, \dots, v_r$ are a $\mathbb{Z}$ -basis of $\mathcal{L}(A(q))$.*

(ii) *If $c = \sum_{j=1}^r z_j v_j$, then, for every j ,*

$$q_j |z_j| \leq d(c). \quad (3.3)$$

(iii) *For every real $H \geq 0$, the subgroup generated by relations of degree at most H is exactly*

$$G_H := \langle c \in \mathcal{L}(A(q)) : d(c) \leq H \rangle_{\mathbb{Z}} = \bigoplus_{q_j \leq H} \mathbb{Z} v_j. \quad (3.4)$$

(iv) *If $q_{(1)} \leq \dots \leq q_{(r)}$ is the sorted list of the radices, then*

$$(\lambda_1(A(q)), \dots, \lambda_r(A(q))) = (2q_{(1)}, \dots, 2q_{(r)}). \quad (3.5)$$

The upper bounds are supplied by the explicit carries. The substantive point is to rule out short integer combinations that could create extra independent directions earlier. The coordinate estimate does precisely that.

3.1 The integral basis

Lemma 3.2. *The carries form a $\mathbb{Z}$ -basis, and their coordinates are given by*

$$z_j = -\frac{1}{a_{j+1}} \sum_{i=1}^j a_i c_i. \quad (3.6)$$

Proof. Each v_j has coordinate sum zero and weighted sum $-q_j a_j + a_{j+1} = 0$, so lies in the lattice. Its highest nonzero coordinate is $j+1$, with coefficient one. Reading from the highest coordinate down shows that the v_j are independent.

For an arbitrary $c \in \mathcal{L}(A(q))$, subtract $c_{r+1} v_r$. This kills coordinate $r+1$ without changing the lattice constraints. Continue backwards, killing coordinate $j+1$ with an integer multiple of v_j . The remainder is supported on coordinates zero and one. Since $a_0 = 0$ and $a_1 = 1$, its weighted-sum constraint kills coordinate one, and its coordinate-sum constraint then kills coordinate zero. This proves integral, not just rational, spanning.

For the coordinate formula, calculate the partial weighted sum of each basis vector through coordinate j . The result is zero for v_ℓ with $\ell < j$, zero for $\ell > j$, and $-a_{j+1}$ for v_j . Linearity gives (3.6). $\square$

3.2 The inequality excluding hidden short relations

Lemma 3.3 (A cut-coordinate bound). *For every $c \in \mathcal{L}(A(q))$ and every $1 \leq j \leq r$, $q_j |z_j| \leq d(c)$.*

Proof. Set

$$P_j = \sum_{i=1}^j a_i c_i^+, \quad N_j = \sum_{i=1}^j a_i c_i^-.$$

All weights in these sums lie between zero and a_j , while each positive or negative mass is at most $d(c)$. Thus $0 \leq P_j, N_j \leq a_j d(c)$, and consequently

$$a_{j+1} |z_j| = \left| \sum_{i=1}^j a_i c_i \right| = |P_j - N_j| \leq a_j d(c).$$

Division by $a_j > 0$ and $a_{j+1} = q_j a_j$ proves the claim. $\square$

Completion of Theorem 3.1. If $d(c) \leq H$ and $q_j > H$, the bound forces $z_j = 0$. Hence every generator of G_H lies in the right side of (3.4). Conversely, $d(v_j) = q_j$, so every basis vector on that right side is itself one of the allowed generators. This proves equality of subgroups. Their rational rank is

$$\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{c : d(c) \leq H\} = \#\{j : q_j \leq H\}. \quad (3.7)$$

The definition of successive minima now gives (3.5). $\square$

Corollary 3.4 (Exact admissibility classification). *For $r \geq 1$, a nondecreasing list $(\ell_1, \dots, \ell_r)$ is the full successive- L^1 -minimum list of the coefficient lattice of a set of $r+2$ distinct integers if and only if every ℓ_j is even and $\ell_1 \geq 4$. For $\ell_j = 2h_j$, one witness is*

$$A = \{0, 1, h_1, h_1 h_2, \dots, h_1 \cdots h_r\}.$$

Proof. Necessity is Proposition 2.2; sufficiency is Theorem 3.1 with $q_j = h_j$. $\square$

For $r = 0$, the empty minimum list is realized by $\{0, 1\}$, whose coefficient lattice is zero. This is the only rank-zero convention used.

4 Rigidity, short-vector enumeration, and a second lower-bound argument

4.1 Only the carry vectors introduce a direction at its first level

Let

$$G_{<Q} = \langle c \in \mathcal{L}(A(q)) : d(c) < Q \rangle_{\mathbb{Z}} = \bigoplus_{q_j < Q} \mathbb{Z}v_j.$$

The filtration theorem is stronger than a list of minima. In fact, the vectors that first leave one of its layers are completely rigid.

Proposition 4.1 (Entrance rigidity). *For any radix value Q that occurs in the list,*

$$\{c \in \mathcal{L}(A(q)) : d(c) \leq Q, c \notin G_{<Q}\} = \{v_j, -v_j : q_j = Q\}. \quad (4.1)$$

Proof. Take a vector on the left. Some basis coordinate z_j is nonzero with $q_j \geq Q$. The inequalities $q_j |z_j| \leq d(c) \leq Q$ imply $q_j = Q$, $|z_j| = 1$ and $d(c) = Q$. Change the sign so that $z_j = 1$. In the notation of Lemma 3.3, $N_j - P_j = Qa_j$, while $N_j \leq Qa_j$ and $P_j \geq 0$. Therefore $N_j = Qa_j$ and $P_j = 0$.

Equality in $N_j \leq a_j \sum_{i=1}^j c_i^- \leq a_j d(c)$, with strictly increasing positive weights, forces all negative mass to be at coordinate j : $c_j = -Q$. Every positive nonzero-weight coordinate must then be beyond j . Their total weighted sum is $Qa_j = a_{j+1}$. The smallest available weight is a_{j+1} , so exactly one unit of positive mass occurs there, and none farther to the right. The other $Q - 1$ units of positive mass are at coordinate zero. Thus $c = v_j$. The reverse inclusion follows directly from the basis and its degrees. $\square$

Corollary 4.2 (All minimizing bases). *Let $m_Q = \#\{j : q_j = Q\}$. Every ordered independent list $c_1, \dots, c_r$ satisfying $\|c_i\|_1 = \lambda_i(A(q))$ consists of the carry vectors, with arbitrary signs and arbitrary permutations among equal radices. Every such list is a $\mathbb{Z}$ -basis. The number of these ordered lists is*

$$2^r \prod_Q m_Q!. \quad (4.2)$$

Proof. At the smallest degree, Proposition 4.1 gives exactly the signed carries at that degree. Independence forces the whole block to use each direction once. Induct over the distinct degree values. Previous blocks span $G_{<Q}$; an independent vector of degree Q must leave that space, so the proposition applies again. Lemma 3.2 and elementary sign and permutation counting finish the proof. $\square$

4.2 An exact, bounded search in carry coordinates

For $r \geq 1$, the norm can be written explicitly as

$$2 \, d \left(\sum_j z_j v_j \right) = \left| \sum_{j=1}^r (q_j - 1) z_j \right| + q_1 |z_1| + \sum_{j=2}^r |z_{j-1} - q_j z_j| + |z_r|. \quad (4.3)$$

A degree- H ball can therefore be enumerated by trying precisely the integer box

$$|z_j| \leq \left\lfloor \frac{H}{q_j} \right\rfloor \quad (1 \leq j \leq r) \quad (4.4)$$

and retaining the vectors that pass (4.3). At most $\prod_j (2 \lfloor H/q_j \rfloor + 1)$ candidates need to be tried. This is an exact finite algorithm, not a claim of polynomial complexity in r . The box is an *outer bound*, not a description of the norm ball. Similarly, G_H in (3.4) is a subgroup generated by a ball; it is not the ball itself and contains vectors of arbitrarily large norm as soon as it is nonzero.

4.3 A short alternative proof when the radices are ordered

The original realization claim can also be checked without the coordinate formula. Suppose $q_1 \leq \dots \leq q_r$, take $0 \neq c \in \mathcal{L}(A(q))$, and let m be its highest nonzero coordinate. Necessarily $m \geq 2$. Change the sign so that $c_m < 0$. The weighted negative part is at least a_m , while the weighted positive part is at most $d(c)a_{m-1}$. Since those weighted sums are equal,

$$d(c) \geq a_m/a_{m-1} = q_{m-1}. \quad (4.5)$$

Thus relations of degree less than q_j are supported on coordinates $0, \dots, j$. The relation space on these $j+1$ coordinates has dimension $j-1$, so it cannot contain j independent vectors. This gives $\lambda_j \geq 2q_j$; the first j carries give the reverse inequality. Unlike the cut-coordinate proof, this particular support argument uses ordered radices.

5 Mixed-radix normal forms and all iterated sumsets

Put

$$P = a_{r+1} = \prod_{j=1}^r q_j, \quad D = \sum_{j=1}^r (q_j - 1), \quad \mathcal{D} = \prod_{j=1}^r \{0, 1, \dots, q_j - 1\}. \quad (5.1)$$

For a digit tuple $d = (d_1, \dots, d_r) \in \mathcal{D}$, write $s(d) = \sum_j d_j$. Let hA denote sums of exactly h elements of A , with repetition allowed, and let $0A = \{0\}$.

Lemma 5.1 (Unique digits and minimum number of summands). *Every integer $n \geq 0$ has a unique expansion*

$$n = tP + \sum_{j=1}^r d_j a_j, \quad t \geq 0, \quad d \in \mathcal{D}. \quad (5.2)$$

The minimum number of nonzero elements of $A(q)$ needed to represent n is $t + s(d)$. Consequently,

$$n \in hA(q) \iff t + s(d) \leq h. \quad (5.3)$$

Proof. Repeated Euclidean division by $q_1, q_2, \dots, q_r$ gives the digits and then the quotient t . Conversely, reduction modulo q_1 determines d_1 ; subtraction and division determine d_2 , and so on. This proves uniqueness.

Start with any representation of n by nonzero elements of $A(q)$. Beginning at a_1 , replace every q_j copies of a_j by one a_{j+1} . Process indices in increasing order. Later replacements never reintroduce a smaller denomination, and each replacement reduces the number of nonzero summands by $q_j - 1$. The final representation has bounded digits and so is the unique expansion (5.2). It follows that its $t + s(d)$ summands are minimal. Zero summands may then be added to obtain exactly h terms, proving (5.3). $\square$

No general assertion that a greedy coin algorithm is optimal is being used: optimality was proved for this divisibility chain by explicit carries. In particular, there is no floating-point comparison and no search in the membership test.

Theorem 5.2 (Exact sumset enumeration). *For every integer $h \geq 0$,*

$$|hA(q)| = \sum_{d \in \mathcal{D}} \max\{0, h - s(d) + 1\}. \quad (5.4)$$

The ordinary generating function is

$$\sum_{h \geq 0} |hA(q)| z^h = \frac{\prod_{j=1}^r (1 + z + \dots + z^{q_j-1})}{(1-z)^2} = \frac{\prod_{j=1}^r (1 - z^{q_j})}{(1-z)^{r+2}}. \quad (5.5)$$

More precisely, the bivariate enumerator is

$$\sum_{h \geq 0} \sum_{n \in hA(q)} z^h w^n = \frac{\prod_{j=1}^r \left(\sum_{d=0}^{q_j-1} (zw^{a_j})^d \right)}{(1-z)(1-zw^P)}. \quad (5.6)$$

Proof. For each digit tuple, exactly the quotients $0 \leq t \leq h - s(d)$ are allowed by (5.3). Distinct digit/quotient pairs give distinct integers, proving (5.4).

Equivalently, every pair (h, n) on the left of (5.6) is represented uniquely by $d \in \mathcal{D}$, $t \geq 0$, and the number $b = h - t - s(d) \geq 0$ of zero summands. Their weights are z^b , $(zw^P)^t$, and $\prod_j (zw^{a_j})^{d_j}$. Summing these independent choices gives (5.6). Setting $w = 1$ and summing the finite geometric series gives (5.5). All operations are valid as formal power series in z . $\square$

6 Consequences for growth and reconstruction

6.1 A finite inclusion–exclusion formula

For an integer m and nonnegative integer b , use $\binom{m}{b} = 0$ when $m < b$. This is a coefficient-counting convention, not the polynomial continuation of binomial coefficients to negative m . Expanding the numerator of (5.5) gives

$$|hA(q)| = \sum_{S \subseteq \{1, \dots, r\}} (-1)^{|S|} \binom{h - \sum_{j \in S} q_j + r + 1}{r + 1}. \quad (6.1)$$

This uses 2^r terms regardless of the magnitude of P . It is therefore complementary to direct digit enumeration, which has P digit tuples, and to polynomial multiplication, whose output degree is D .

When the sorted radices satisfy $q_{(1)} < q_{(2)}$, the interval $q_{(1)} \leq h < q_{(2)}$ has exactly one active nonempty singleton in (6.1), and hence

$$|hA(q)| = \binom{h+r+1}{r+1} - \binom{h-q_{(1)}+r+1}{r+1}.$$

Below the smallest radix there are no collisions at all. These initial regimes agree with O'Bryant's general Theorem 2 [1]; they are not claimed as new general sumset theorems here. If the smallest radix Q occurs m times, then at $h = Q$ the deficit from $\binom{Q+r+1}{r+1}$ is exactly m .

6.2 Discrete curvature and the exact linearity threshold

Define the digit-sum polynomial

$$C_q(z) = \prod_{j=1}^r (1 + z + \cdots + z^{q_j-1}) = \sum_{s=0}^D p_s z^s. \quad (6.2)$$

Here p_s counts digit tuples of sum s . Every integer s between zero and D occurs: this follows inductively because the sum of two integer intervals is an integer interval. Thus $p_s > 0$ throughout that interval. The involution $d_j \mapsto q_j - 1 - d_j$ gives $p_s = p_{D-s}$. Moreover,

$$\sum_s p_s = P, \quad \sum_s s p_s = \frac{PD}{2}. \quad (6.3)$$

The second identity follows either by the same involution or by summing one digit coordinate at a time. In particular, PD is even.

Writing $N(h) = |hA(q)|$ and $N(-1) = N(-2) = 0$, equation (5.5) yields

$$N(h) - 2N(h-1) + N(h-2) = p_h \quad (h \geq 0), \quad (6.4)$$

where $p_h = 0$ for $h > D$. The first differences increase to P , and their successive increases are the symmetric digit-sum distribution.

Theorem 6.1 (Sharp onset of affine growth). *If $r \geq 1$, then for every $h \geq D-1$,*

$$|hA(q)| = P(h+1) - \frac{PD}{2}. \quad (6.5)$$

The least nonnegative index from which this formula holds permanently is $D-1$. If $D \geq 2$, its failure immediately before that index is exactly

$$|(D-2)A(q)| = P(D-1) - \frac{PD}{2} + 1. \quad (6.6)$$

Proof. In (5.4), once $h \geq D-1$ every term $h - s(d) + 1$ is nonnegative; the possible boundary term at $s(d) = D$ is zero. Remove the maximum and use (6.3) to obtain (6.5). At $h = D-2$, precisely one digit tuple has $s(d) = D$, and its untruncated contribution would be -1 . All other untruncated contributions are nonnegative. Truncation therefore adds exactly one, giving (6.6). If $D = 1$, the formula already holds at $h = 0$, which is the least allowed index. $\square$

For $r = 0$ the separate convention is $N(h) = h+1$, valid for all $h \geq 0$. The denominator $(1-z)^2$ in the first fraction of (5.5) is reduced: its numerator takes the nonzero value P at $z = 1$.

6.3 Permuting the radices and recovering their multiset

Although the order of the radices changes the intermediate elements of $A(q)$, it changes neither the successive minima nor any sumset size. The first assertion is Theorem 3.1; the second is the symmetric product in (5.5). The actual sumsets can change, as can the coefficients in the carry basis.

Proposition 6.2 (Reconstruction within the chain family). *The complete sequence $(|hA(q)|)_{h \geq 0}$ determines the multiset of radices of a multiplicative-chain set.*

Proof. The value at $h = 1$ determines $r = |A(q)| - 2$. Multiply the generating series by $(1 - z)^{r+2}$ to recover $F(z) = \prod_j (1 - z^{q_j})$. If $F \neq 1$, the least positive exponent with nonzero coefficient is the smallest remaining radix m , and its coefficient is minus the multiplicity of m . Remove the corresponding factor $(1 - z^m)$ to that multiplicity and repeat. The process terminates after the known number r of factors. $\square$

This is not an inverse theorem for arbitrary integer sets: membership in the chain family is essential to the reconstruction argument.

7 The associated binomial ideal

The sumset proof is elementary and independent of this section. An algebraic formulation nevertheless explains why its generating function has such a simple product numerator.

Let K be any field, and put $R = K[x_0, \dots, x_{r+1}]$. Give each x_i ordinary degree one. Define

$$\varphi : R \longrightarrow K[u, t], \quad x_i \longmapsto ut^{a_i}.$$

The degree- h image has one basis monomial $u^h t^n$ for each $n \in hA(q)$, so its dimension is $|hA(q)|$. For $1 \leq j \leq r$ set

$$f_j = x_j^{q_j} - x_0^{q_j-1} x_{j+1}. \quad (7.1)$$

Theorem 7.1 (Carry presentation). *The kernel of φ is the ideal $(f_1, \dots, f_r)$. A K -basis of the quotient consists of the monomials*

$$x_0^b x_{r+1}^t \prod_{j=1}^r x_j^{d_j}, \quad b, t \geq 0, \quad 0 \leq d_j < q_j. \quad (7.2)$$

For lexicographic order $x_1 > x_2 > \dots > x_{r+1} > x_0$, the f_j form a Gröbner basis with leading monomials $x_j^{q_j}$. They are also a regular sequence.

Proof. Every f_j lies in the kernel because $q_j a_j = a_{j+1}$. Replacing $x_j^{q_j}$ by $x_0^{q_j-1} x_{j+1}$, in increasing order of j , reduces every monomial to one of (7.2). The replacement preserves the ordinary degree and its weighted exponent. Distinct normal monomials have distinct images: equality of their t -exponents gives equality of their digits and top quotients by Lemma 5.1, and equality of their u -exponents then gives equality of b .

A polynomial in $\ker \varphi$ reduces modulo the f_j to a linear combination of normal monomials. Their distinct image monomials are linearly independent in $K[u, t]$, so every coefficient in that remainder is zero. This proves both the kernel assertion and the quotient basis.

In the stated lexicographic order, each carry replacement reduces a monomial: its first changed variable is x_j and its exponent decreases. Division by the f_j therefore produces a remainder containing no $x_j^{q_j}$. The kernel argument says that an element of the ideal has zero remainder, which is the Gröbner-basis property. It also follows from the fact that the leading monomials are pairwise relatively prime, but that criterion is not needed here.

For regularity, impose the equations in the order $f_1, f_2, \dots$. After imposing $f_1, \dots, f_m$, the quotient is a free module over $K[x_0, x_{m+1}, \dots, x_{r+1}]$, with basis $\prod_{j=1}^m x_j^{d_j}$, $0 \leq d_j < q_j$. This follows inductively from division by the monic polynomial $x_m^{q_m} - x_0^{q_m-1} x_{m+1}$. The next polynomial f_{m+1} is a nonzero element of that polynomial base ring, an integral domain. Multiplication by it is injective on the free module. Thus it is not a zero divisor after the preceding equations. All the polynomials have positive degree, so the resulting ideal is proper. This proves regularity. $\square$

Thus the image ring is a complete intersection in the concrete sense of a polynomial ring modulo the displayed regular sequence. It is a free module of rank P over $K[x_0, x_{r+1}]$. Counting the degrees of the normal monomials recovers (5.5) without invoking any general formula for Hilbert series. In particular, we did not infer ideal generation merely from lattice-basis generation; that inference is not valid in general.

8 Examples and limits of the construction

8.1 A five-element example

For $q = (2, 3, 5)$, the construction is

$$A = \{0, 1, 2, 6, 30\},$$

with carry basis

$$\begin{pmatrix} 1 & -2 & 1 & 0 & 0 \\ 2 & 0 & -3 & 1 & 0 \\ 4 & 0 & 0 & -5 & 1 \end{pmatrix}.$$

The successive norms are 4, 6, 10. At degree bounds $H < 2$, $2 \leq H < 3$, $3 \leq H < 5$, and $H \geq 5$, the ranks generated by short relations are, respectively, 0, 1, 2, 3. Here

$$C_q(z) = 1 + 3z + 5z^2 + 6z^3 + 6z^4 + 5z^5 + 3z^6 + z^7,$$

and the sumset sizes begin

$$1, 5, 14, 29, 50, 76, 105, 135, 165, \dots$$

Since $P = 30$ and $D = 7$, the exact affine formula is $|hA| = 30h - 75$ for $h \geq 6$; at $h = 5$ it underestimates by exactly one. The bivariate formula distinguishes the actual sums, while this ordinary sequence records only their number.

8.2 Repeated minima and large gaps

For $q = (2, 2, 4, 7)$,

$$A = \{0, 1, 2, 4, 16, 112\}, \quad (\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (4, 4, 8, 14).$$

Exactly two independent directions enter at degree two, but no direction enters at degree three. The number of ordered successive minimizing bases is $2^4 \cdot 2! = 32$. Replacing 7 by any larger integer postpones the last independent relation to that degree without changing the earlier minima.

For $q = (4, 9)$, the witness is $\{0, 1, 4, 36\}$ with successive norms 8, 18. Inclusion–exclusion gives, for every $h \geq 0$,

$$|hA| = \binom{h+3}{3} - \binom{h-1}{3} - \binom{h-6}{3} + \binom{h-10}{3}.$$

The zero convention on binomial coefficients is essential for small h . This example gives two independently chosen thresholds by the same construction that works for an arbitrarily long list.

8.3 Equal minima need not determine equal sumset sizes

Consider

$$A = \{0, 1, 3, 9\}, \quad B = \{0, 1, 5, 7\}.$$

Both have successive norms $(6, 6)$. This follows for A from the theorem. For B , its ten unordered two-term sums are

$$0, 1, 2, 5, 6, 7, 8, 10, 12, 14,$$

so there is no nonzero relation of degree at most two. On the other hand,

$$(-2, 2, 1, -1), \quad (0, 1, -3, 2)$$

are independent coefficient relations of degree three. This proves its minimum list without assuming any unproved classification.

Nevertheless, $|4A| = 27$ while $|4B| = 26$. The first value follows from (5.5). To check the second directly, take l summands from $\{5, 7\}$ and fill the remaining $4 - l$ positions with zeros and ones. For $l = 0, 1, 2, 3, 4$ the resulting sets are, respectively,

$$[0, 4], \quad [5, 10], \quad [10, 16], \quad [15, 22], \quad \{20, 22, 24, 26, 28\}.$$

Their union is $[0, 22] \cup \{24, 26, 28\}$, of size 26. Thus even the complete successive-minimum list does not specify a general sumset sequence.

8.4 The diameter is controlled, not optimized

The explicit witness has diameter $P = \prod_j h_j$. This can be much larger than necessary. If all r half-minima are two, the interval

$$I = \{0, 1, \dots, r + 1\}$$

also works: its r independent second-difference relations $e_{j-1} - 2e_j + e_{j+1}$ have norm four, the universal lower bound. Every $(r + 2)$ -element integer set has diameter at least $r + 1$, so this interval has minimum possible diameter for that list. The chain witness instead has diameter 2^r .

There is also an elementary general lower bound. Suppose $k = r + 2$ and the first half-minimum is h_1 . At $s = h_1 - 1$, all unordered s -term sums are distinct; otherwise their difference would be a relation of degree at most s . A set of diameter M has sA in an integer interval of length sM . Therefore

$$M \geq \max \left\{ k - 1, \left\lceil \frac{\binom{h_1+k-2}{k-1} - 1}{h_1 - 1} \right\rceil \right\}. \quad (8.1)$$

The theorem provides an upper bound $\prod_j h_j$, not an assertion that it matches this lower bound or solves diameter minimization.

9 Algorithms, exact checks, and proof audit

9.1 Constructing and using a witness

The witness is built by r integer multiplications. If $L = \sum_j \lceil \log_2 q_j \rceil$, every positive element has at most $L + 1$ binary digits, and the entire list has $O(rL)$ bits. Even elementary integer multiplication therefore gives a polynomial-time construction in the bit length of the input list. This is an existence algorithm; it does not solve a shortest-vector problem for an arbitrary lattice.

Membership in $hA(q)$ requires r Euclidean divisions to get the digits, followed by checking $t + s(d) \leq h$. A canonical length- h representation, when it exists, is

$$(h - t - s(d), d_1, \dots, d_r, t).$$

Computing the whole digit-sum polynomial can be done using a sliding-window convolution by $1 + z + \dots + z^{q_j-1}$ at each stage, requiring $O(r(D+1))$ integer additions or subtractions. Formula (5.4) then evaluates any $N(h)$ in $O(D+1)$ operations, or (6.5) applies immediately beyond the exact threshold. For small rank and very large radices, (6.1) avoids creating a degree- D polynomial.

9.2 What the verifier enumerates independently

The main verifier does not infer minima from the claimed answer. For a set A and a bound H , it enumerates every weak composition $(m_0, \dots, m_{k-1})$ of H and groups them by their weighted sum. For each group, it subtracts one representative from every other member. These differences span exactly the degree-at-most- H relation space. Indeed, any relation of degree $d \leq H$ can have its positive and negative parts padded by $H-d$ zeros, so it appears as a difference in a group; any difference of two group members is the difference of their two representative differences. The rank is computed by fraction-free integer Gaussian elimination, not by floating-point singular values.

The recorded checks are summarized below. Overlap between the ordered and arbitrary-order profile collections is intentional; their counts are not advertised as disjoint cases.

Check	Recorded count
Nondecreasing lists, $1 \leq r \leq 4$, $2 \leq q_j \leq 6$	125
Degree/rank checks for those lists	754
Weak compositions enumerated for those lists	56,493
Collision generators checked for those lists	10,943
Ordered tuples in every order, $1 \leq r \leq 4$, $2 \leq q_j \leq 5$	340
Degree/rank checks for those tuples	1,886
Weak compositions enumerated for those tuples	109,392
Collision generators checked for those tuples	22,909
Independent repeated-set-addition profiles	34
Exact sumset-size checks by that method	346
Individual membership tests against those sumsets	134,080
Exhaustive vector-ball comparisons using all multiset pairs	5
Vectors in those five balls, including their zeros	79

Every recorded check passed. Additional examples include rank zero, repeated minima, widely separated minima, and the two non-equivalent sumset sequences in Section 8. All computations use Python integers and the standard library; no external symbolic service or random sampling of relations is involved.

In the five ball checks, every pair of representations in every bucket is used, not only differences to a representative. This independently gives the entire finite relation ball and is compared with the bounded carry-coordinate enumerator. Entrance rigidity is checked on those balls. The finite tests supplement the proof; they do not replace any quantifier in Theorem 3.1.

9.3 The principal logical hazards

A list of r relations with the desired lengths proves only upper bounds on the successive minima. Lemma 3.3 supplies the missing lower bounds for every integer combination. Triangular elimination separately establishes integral generation, rather than merely full rank. Ties are handled by the exact rank profile, with no assumption that the minima are strictly increasing.

The normal-form proof counts integers only after uniqueness is established. The ideal proof separately establishes kernel equality, which does not follow from a lattice basis alone. Finally, affine growth starts at $D-1$, not merely at the sufficient but non-sharp bound D .

10 Conclusion

The realization problem has an explicit all-rank answer. Prescribed half-minima $h_j \geq 2$ can be installed as successive multiplicative ratios in a set containing zero and one. The carry relations furnish the upper bounds, and the partial weighted sums furnish simultaneous lower bounds on every carry coordinate. There are no additional arithmetic obstructions on the list of minima.

The same construction supports more structure than the existence theorem requires: its short-relation filtration is a direct-summand filtration, its minimizing bases are rigid up to signs and ties, and its sumsets admit unique digit normal forms. The resulting product generating function and sharp affine onset are consequences of those normal forms. None of these claims depends on an unproved general principle about arbitrary lattices or arbitrary sumsets.

The mathematical proof and the bibliographic status are separate matters. The former is fully given here. The latter remains a research-draft claim of a resolution of the conjecture as located in the inspected source; independent scrutiny and a broader priority check remain appropriate.

A The randomized area catalogue

The catalogue was fixed before the draw. The call was `secrets.randbelow(96)`; its recorded zero-based output was 0. Thus the chosen area is entry 1. The saved result, rather than a seeded replay, is the audit record: operating-system entropy is not reproducible from a seed that was never used. The archived script defaults to displaying the existing record and will not overwrite it with a fresh draw.

- | | | |
|--|---------------------------------------|---------------------------------------|
| 1. Additive combinatorics | 32. Combinatorial commutative algebra | 64. Diophantine approximation |
| 2. Algebraic combinatorics | 33. Invariant theory | 65. Diophantine equations |
| 3. Analytic combinatorics | 34. Finite group theory | 66. Multiplicative number theory |
| 4. Extremal graph theory | 35. Combinatorial group theory | 67. Partition theory |
| 5. Structural graph theory | 36. Semigroup theory | 68. q-series |
| 6. Graph coloring | 37. Nonassociative algebra | 69. Special functions |
| 7. Spectral graph theory | 38. Lie algebras | 70. Orthogonal polynomials |
| 8. Hypergraph theory | 39. Representation theory | 71. Approximation theory |
| 9. Ramsey theory | 40. Category theory | 72. Potential theory |
| 10. Design theory | 41. Type theory | 73. Harmonic analysis |
| 11. Finite geometry | 42. Universal algebra | 74. Fourier analysis on finite groups |
| 12. Matroid theory | 43. Order theory | 75. Functional equations |
| 13. Discrete geometry | 44. Lattice theory | 76. Matrix theory |
| 14. Convex geometry | 45. Computability theory | 77. Nonnegative matrices |
| 15. Geometry of numbers | 46. Proof theory | 78. Tensor algebra |
| 16. Lattice polytopes and Ehrhart theory | 47. Model theory | 79. Polynomial inequalities |
| 17. Tiling theory | 48. Set-theoretic topology | 80. Moment problems |
| 18. Combinatorial game theory | 49. General topology | 81. Optimization |
| 19. Permutation patterns | 50. Knot theory | 82. Discrete optimization |
| 20. Combinatorics on words | 51. Low-dimensional topology | 83. Numerical analysis |
| 21. Automata and formal languages | 52. Algebraic topology | 84. Difference equations |
| 22. Symbolic dynamics | 53. Persistent homology | 85. Differential algebra |
| 23. Substitution dynamical systems | 54. Metric geometry | 86. Differential equations |
| 24. Ergodic theory | 55. Graph limits | 87. D-finite functions |
| 25. Topological dynamics | 56. Probabilistic combinatorics | 88. Integer sequences |
| 26. Arithmetic dynamics | 57. Discrete probability | 89. Enumerative topology |
| 27. Complex dynamics | 58. Markov chains | 90. Geometric group theory |
| 28. Real dynamical systems | 59. Random walks | 91. Computational geometry |
| 29. Numerical semigroups | 60. Information theory | 92. Boolean functions |
| 30. Commutative algebra | 61. Coding theory | 93. Algebraic statistics |
| 31. Monomial ideals | 62. Finite fields | 94. Tropical geometry |
| | 63. Elementary number theory | 95. Real algebraic geometry |
| | | 96. Incidence geometry |

B Reproduction and package contents

Run the finite checks from the package root with

```
python3 code/verify.py
```

The program regenerates the CSV and JSON outputs under `data/`. The two main mathematical routines can also be used directly:

```
# Add this package's code/ directory to the Python import path:
import sys
sys.path.insert(0, "code")
from mixed_radix import construct, carry_basis, sumset_size
from mixed_radix import canonical_representation

q = (2, 3, 5)
assert construct(q) == (0, 1, 2, 6, 30)
assert sumset_size(q, 6) == 105
assert canonical_representation(q, 4, 35) == (0, 1, 2, 0, 1)
```

Python also has a standard-library module named `code`; avoid using that name as a package import. The supported import is `from mixed_radix import ...` after adding the directory.

Compile the article using `latexmk -pdf article.tex`, or run `pdflatex article.tex` twice. The package includes the complete L^AT_EX source, this PDF, the implementation, the verifier, the recorded area choice, exact results, and separate source and proof audits. No source paper PDF, font files, or external dependencies are redistributed.

References

- [1] Kevin O'Bryant, *On Nathanson's Triangular Number Phenomenon*, arXiv:2506.20836v2, 18 August 2025. <https://arxiv.org/abs/2506.20836v2>. The target is the unnumbered conjecture after Lemma 1, Section 1.1, page 2. Definitions and the adjacent statement were checked in both the HTML and the displayed PDF page on 20 September 2026.
- [2] Melvyn B. Nathanson, *Explicit sumset sizes in additive number theory*, Canadian Mathematical Bulletin, published online 4 December 2025; preprint arXiv:2505.05329. <https://arxiv.org/abs/2505.05329>. Cited for the surrounding range-of-sumset-sizes problem, not as an ingredient in the proofs of this article.